

Jakarta Enterprise Edition

(Jakarta EE)

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Rodrigo Martins Pagliares

Introduction to Jakarta EE

Overview of Jakarta EE

Lecture: Overview of Jakarta EE

Module: Introduction to Jakarta EE

Jakarta EE

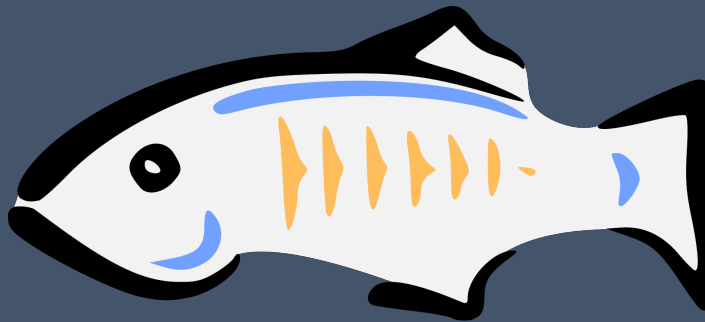
- Set of specifications, for enterprise features such as distributed computing and web services.
 - ✓ The specifications extend Java SE.
- Formerly known as Java 2 Platform, Enterprise Edition (J2EE), and Java Platform, Enterprise Edition (Java EE).
- The Jakarta EE specification defines APIs and their interactions.



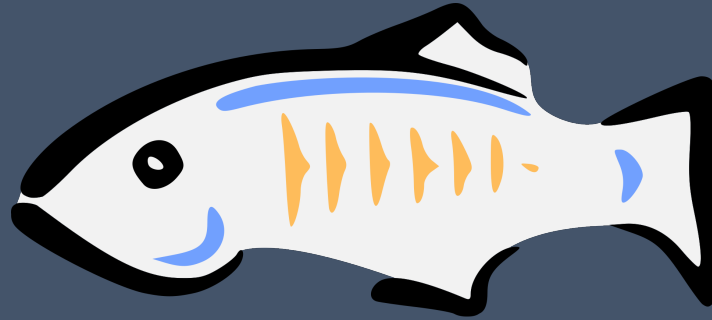
Jakarta EE reference runtimes

- Jakarta EE applications are run on reference runtimes.
- Reference runtimes are microservices or application servers, which handle transactions, security, scalability, concurrency and management of the components it is deploying.
- Providers of reference runtimes must meet certain conformance requirements in order to declare their products as Jakarta EE compliant.

Certified Jakarta EE 8 application servers



Certified Jakarta EE 9 application server



Glassfish

* In Jan 2021, only **Glassfish** was fully Jakarta EE certified (full profile and web profile)

Jakarta® EE

Open Source Cloud Native Java

Powered by participation, Jakarta EE is focused on enabling community-driven collaboration and open innovation for the cloud. Build modern and portable enterprise applications and protect your investments in Java EE.

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Specifications

Jakarta EE Platform

Defines a platform for hosting Jakarta EE applications

Jakarta EE Web Profile

Web profile of platform for hosting Jakarta EE applications

Jakarta Activation

Defines standard services for MIME type data access

Jakarta Annotations

Declarative style annotations for common semantic concepts

Jakarta Authentication

Low-level container SPI for authentication mechanisms

Jakarta Authorization

Low-level container SPI for authorization modules

Jakarta Batch

API plus an XML-based batch job specification language

Jakarta Bean Validation

Object level constraint declaration and validation facility

Jakarta Concurrency

API for app level concurrency that integrates with container

Jakarta Connectors

Jakarta Contexts and Dependency Injection

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- [Specification Process](#)

<https://jakarta.ee/specifications>

VERSION 9 NEW RELEASE

JAKARTA EE 9

AN OPEN, COMMUNITY-DRIVEN PLATFORM
FOR FUTURE INNOVATION OF
ENTERPRISE JAVA TECHNOLOGIES

**DOWNLOAD
COMPATIBLE PRODUCTS**

SPECIFICATIONS

Previous Release: Jakarta EE 8

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JAKARTA EE
FEATURES AND BENEFITS

<https://jakarta.ee/release/9>

Jakarta EE Platform 9

The Jakarta EE Platform defines a standard platform for hosting Jakarta EE applications.

- [Jakarta EE Platform 9 Release Record](#)
 - [Jakarta EE Platform 9 Release Plan](#)
- [Jakarta EE Platform 9 Specification Document \(PDF\)](#)
- [Jakarta EE Platform 9 Specification Document \(HTML\)](#)
- [Jakarta EE Platform 9 Javadoc](#)
- [Jakarta EE Platform 9 TCK\(sig,sha,pub\)](#)
- Maven coordinates
 - [jakarta.platform:jakarta.jakartaee-api:jar:9.0.0](#)

Jakarta EE 9 Schedule

- [Jakarta EE 9 Schedule](#)

Compatible Implementations

- [Eclipse Glassfish 6.0.0 RC2](#)

Ballots

Plan Review

The Specification Committee Ballot concluded successfully on 2020-01-15 with the following results.

Representative	Representative for:	Vote
Kenji Kazumura, Michael DeNicola		
Dan Bandera, Kevin Sutter		
Bill Shannon, Ed Bratt		

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<https://jakarta.ee/specifications/platform/9>

Jakarta EE Tutorial

Jakarta EE Tutorial Project

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About this Project

The Jakarta EE Tutorial Project is the official site for the *Jakarta Platform, Enterprise Edition (Jakarta EE) Tutorial*. The Jakarta EE Tutorial teaches and demonstrates the Jakarta EE features that are used to develop enterprise applications.

[View the Jakarta EE Tutorial](#)

The latest build of the Jakarta EE Tutorial is automatically published to this site as HTML via a Jenkins job that watches for changes to the master branch of the repository. The tutorial is authored in [AsciiDoc](#). AsciiDoc is similar to markdown but is particularly suited for user documentation. If you are interested in forking the tutorial source files or building the tutorials locally, then see the [repository](#).

Related Projects

- [eclipse-ee4j/jakartaee-tutorial-examples](#): this repository contains the example code that is used in the Jakarta EE Tutorial.
- [Your First Cup: An Introduction to the Jakarta EE Platform](#): an introductory tutorial suitable for beginners.
- [eclipse-ee4j/firstcup-examples](#): this repository contains the example code that is used in the Your First Cup tutorial.

How To Contribute

The easiest way to contribute is by

The Jakarta EE Tutorial is open for contributions and your help is greatly appreciated. If you have an idea for the tutorial and want to

Project Resources

- [Sources](#)
- [Documentation](#)
- [Download](#)
- [Issue Tracker](#)
- [Mailing list](#)

<https://eclipse-ee4j.github.io/jakartaee-tutorial>



eclipse-ee4j / cargotracker

Watch

17

Star

79

Fork

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<> Code

Issues 25

Pull requests 1

Actions

Projects

Wiki

Security

Insights

master

4 branches

2 tags

Go to file

Code



m-reza-rahman Update cargo_tracker_soapUI_project.xml

3dbdf22 7 days ago 243 commits



src

Update cargo_tracker_soapUI_project.xml

7 days ago



.gitignore

Fixing warning messages.

5 months ago



LICENSE.md

Update license.

6 months ago



README.md

Update README.md

9 days ago



cargo_tracker_cover.png

Add files via upload

3 months ago



pom.xml

Minor cleanup.

2 months ago

README.md

Eclipse Cargo Tracker - Applied Domain-Driven Design Blueprints for Jakarta EE

Overview

The project demonstrates how you can develop applications with Jakarta EE using widely adopted architectural best practices like Domain-Driven Design (DDD).

About

The project demonstrates how you can develop applications with Jakarta EE using widely adopted architectural best practices like Domain-Driven Design (DDD).

eclipse-ee4j.github.io/cargotracker/

java

ddd

domain-driven-design

javaee

jakartaee

Readme

MIT License

Releases 2

Minor bug fix release (Java... Latest
22 days ago

+ 1 release

Contributors 7

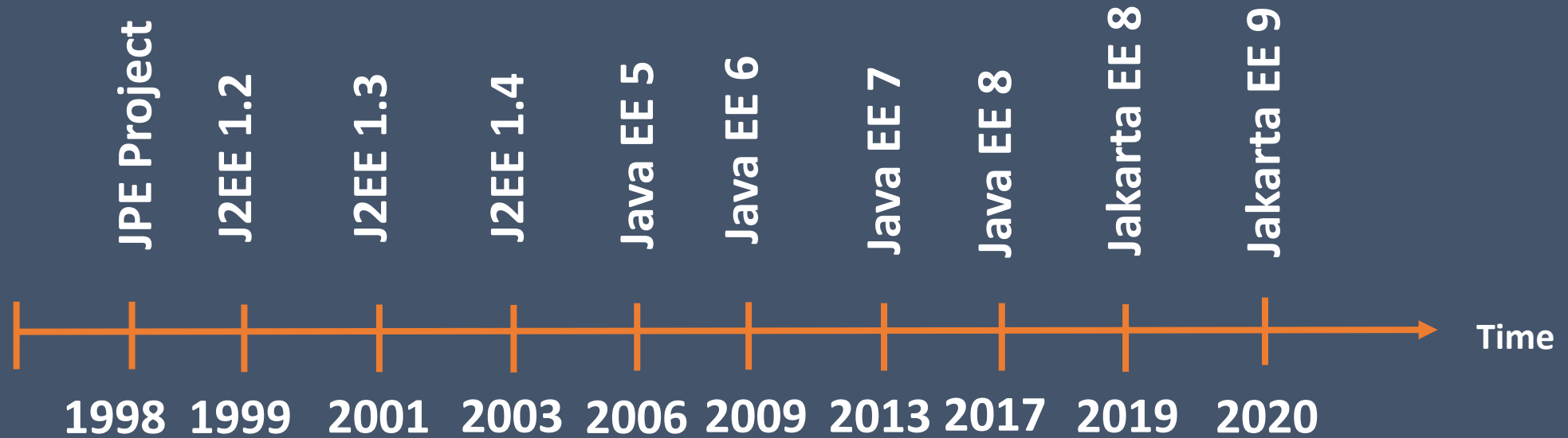
<https://github.com/eclipse-ee4j/cargotracker>

Jakarta EE evolution

Lecture: Jakarta EE evolution

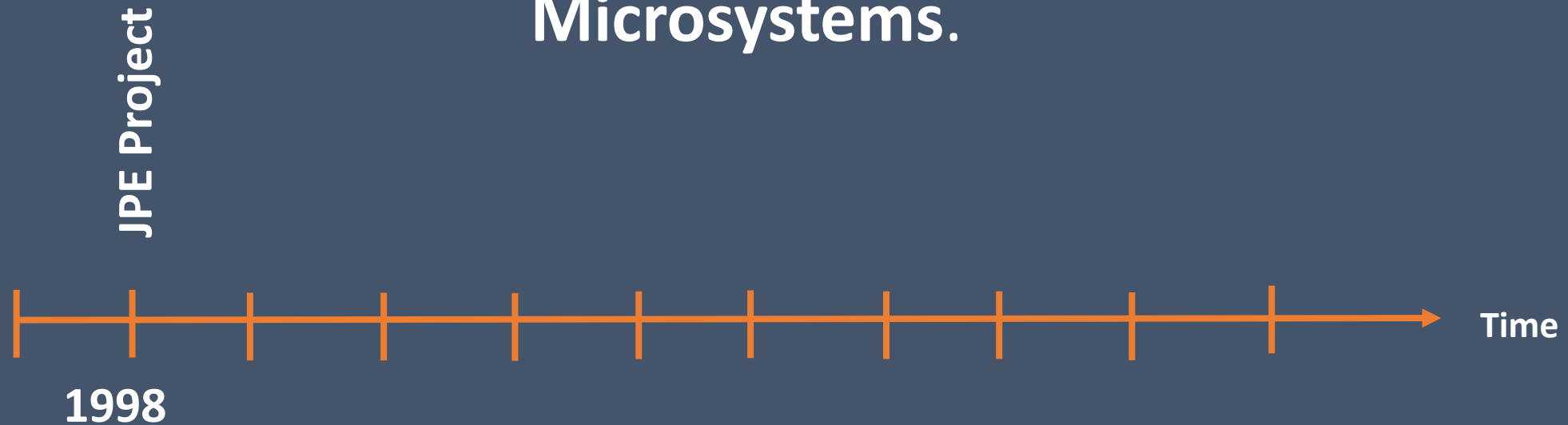
Module: Introduction to Jakarta EE

Jakarta EE evolution



Java Professional Edition (JPE)

Project JPE (Java Professional Edition) announced in **May 1998** at **Sun Microsystems**.



J2EE 1.2

- December **1999**
- Steep **learning curve**
- Very popular
- **Required** a lot of **XML configuration**
- **Specifications:** Servlets, JSP, EJB, JMS, RMI, JDBC, JTA.



J2EE 1.3

- September **2001**.
- Addition and enhancements of existing specifications.
- **Specifications:** JSTL, JAACS, CMP, Connector architecture, ...



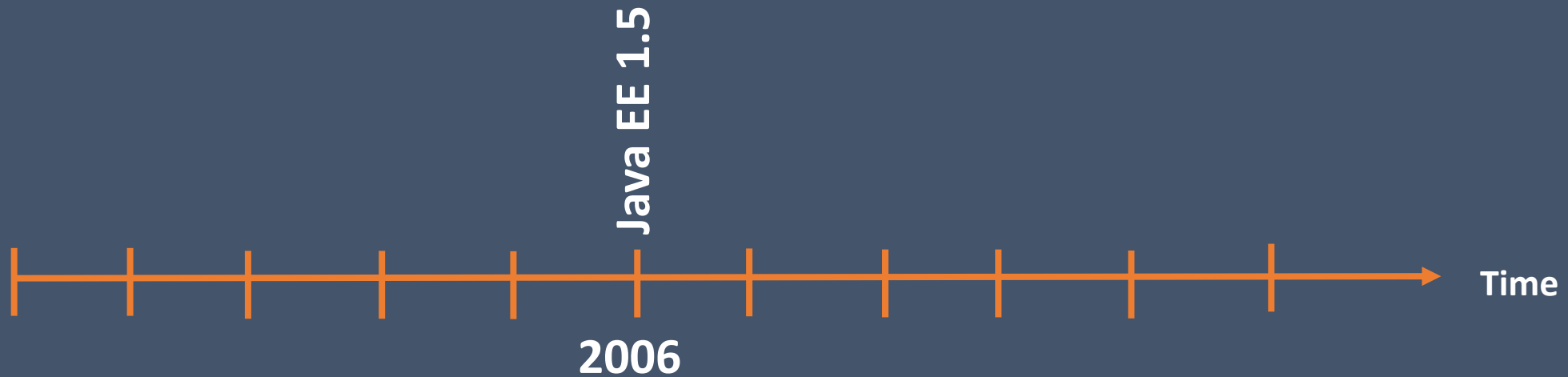
J2EE 1.4

- November 2003
- Steep learning curve
- Many new specifications
- **Specifications:** Web services, management, deployment, JSF, JAXP...



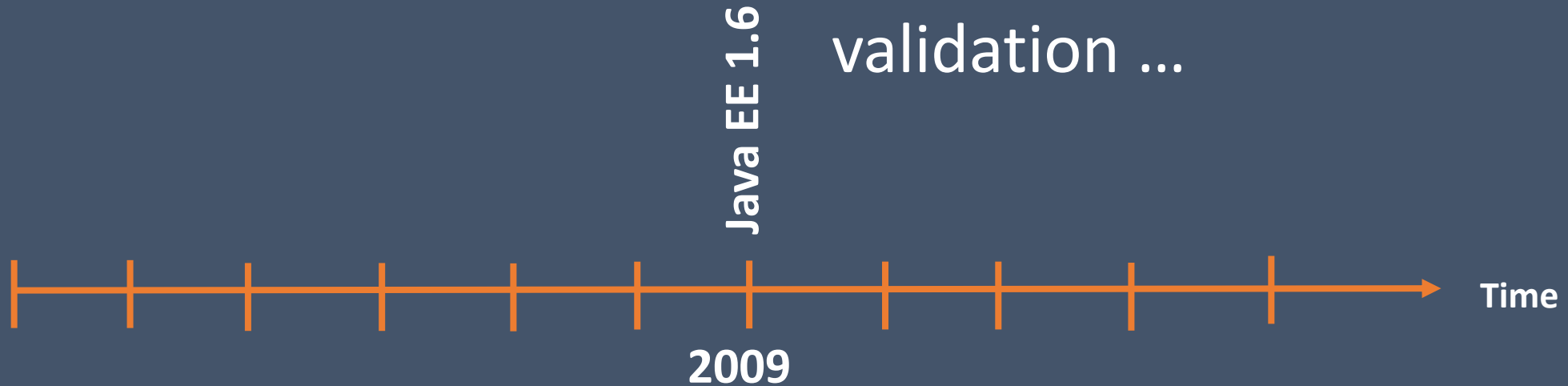
Java EE 1.5

- May 2006.
- J2EE left behind
- Ease of development
- Specifications: Annotations, EJB 3.0, JPA, ...



Java EE 1.6

- December 2009.
- **Flexible.**
- **Specifications:** Extensibility, profiles, EJB lite, RESTful, CDI, web profile, Bean validation ...



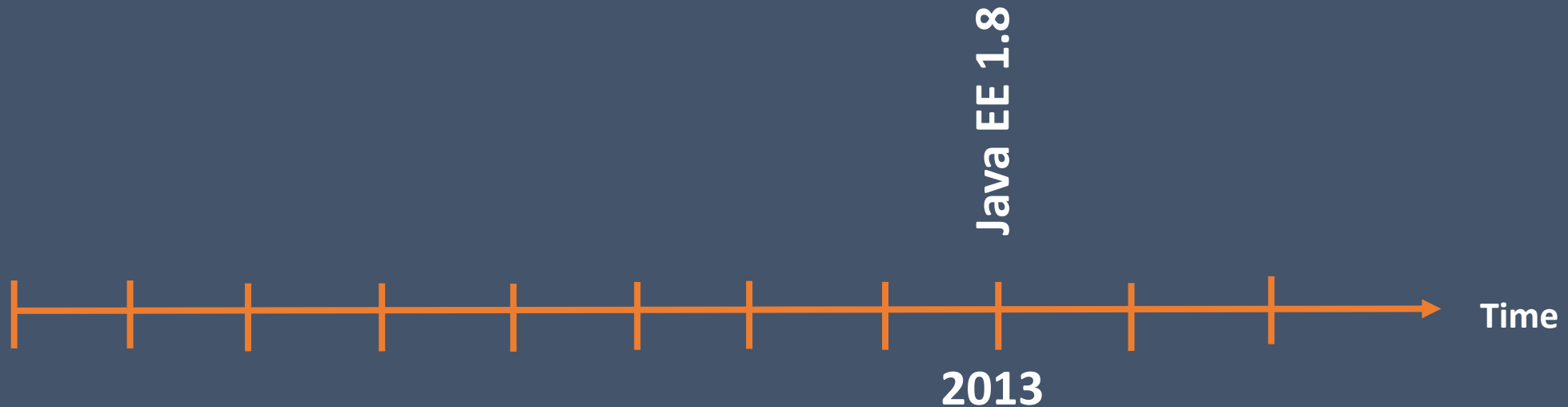
Java EE 1.7

- April **2013**
- **Stable**
- Enhancements in **JSF** and **EJB**
- **Specifications:** WebSockets, JSON-P



Java EE 1.8

- August 2017.
- **Updates** to many **specifications**.
- Oracle announced the intention to open-source Java EE
- Start of transition to **Eclipse Foundation** with the name **Jakarta EE**.



Jakarta EE 8

- September 2019.
- **Fully compatible** with Java EE 8.



Jakarta EE 9

- November **2020**.
- javax.* to jakarta.* namespace change.



Jakarta EE 10?

Jakarta EE 10

*“Ideally, **Jakarta EE 10** will be released by **late 2021** to meet the **community’s long-time request for one major platform release a year**”. (Eclipse foundation).*

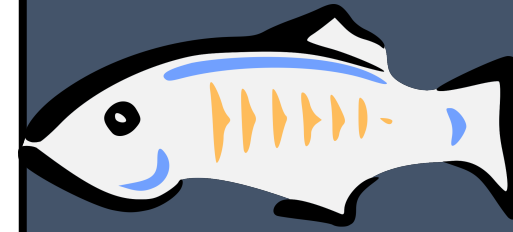
Source: <https://bit.ly/35uoASa>



Jakarta EE 9 profiles

Lecture: Jakarta EE 9 profiles

Module: Introduction to Jakarta EE



Jakarta Authentication 2.0	Jakarta Bean validation 3.0	Jakarta Annotations (JSR-250) 2.0	Jakarta Contexts and Dependency Injection (CDI) 3.0	Jakarta Dependency Injection 2.0	Jakarta Enterprise Beans (EJB) 4.0 lite
Jakarta Debugging Support for other Languages (JSR-45) (EL) 2.0	Jakarta Expression Language (EL) 4.0	Jakarta Interceptors 2.0	Jakarta JSON-Binding 2.0	Jakarta JSON-Processing (JSON-P) 2.0	Jakarta Managed Beans 2.0
Jakarta Security 2.0	Jakarta Server Pages (JSP) 3.0	Jakarta Standard Tag Library (JSTL) 2.0	Jakarta Persistence (JPA) 3.0	Jakarta RESTful Web Services (JAX-RS) 3.0	Jakarta Server Faces 3.0
Jakarta Servlet 5.0	Jakarta Transactions (JTA) 2.0	Jakarta Websocket 2.0	Jakarta Messaging 3.0	Jakarta Activation 2.0	Jakarta Mail 2.0
Jakarta Connectos 2.0	Jakarta Concurrency 2.0	Jakarta Batch 2.0	Jakarta Authorization 2.0	Jakarta Enterprise Beans (EJB) 3.2	Jakarta Enterprise Beans (EJB) 2.x API Group
Jakarta Enterprise Web Services 2.0	Jakarta SOAP with Attachments 2.0	Jakarta Web Services Metadata 3.0	Jakarta XML Web Services 3.0	Jakarta XML Binding 3.0	

Required

Optional

Jakarta EE 9 Full Profile

Jakarta Authentication
2.0

Jakarta Bean
validation 3.0

Jakarta Annotations
(JSR-250) 2.0

Jakarta Contexts and
Dependency Injection
(CDI) 3.0

Jakarta Dependency
Injection 2.0

Jakarta Enterprise
Beans (EJB) 4.0 lite

Jakarta Debugging
Support for other
Languages (JSR-45)
(EL) 2.0

Jakarta Expression
Language (EL) 4.0

Jakarta Interceptors
2.0

Jakarta JSON-Binding
2.0

Jakarta
JSON-Processing
(JSON-P) 2.0

Jakarta Managed
Beans 2.0

Jakarta Security 2.0

Jakarta Server Pages
(JSP) 3.0

Jakarta Standard Tag
Library (JSTL) 2.0

Jakarta Persistence
(JPA) 3.0

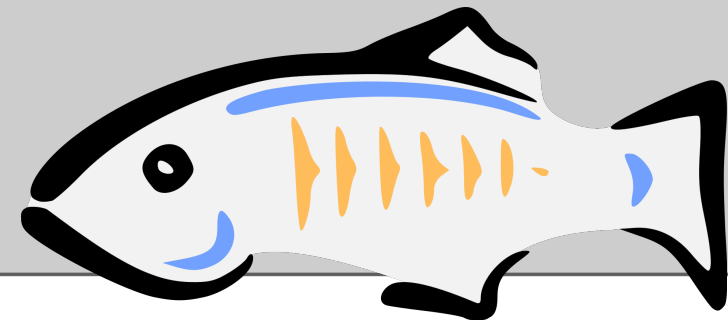
Jakarta RESTful Web
Services (JAX-RS) 3.0

Jakarta Server Faces
3.0

Jakarta Servlet 5.0

Jakarta Transactions
(JTA) 2.0

Jakarta Websocket 2.0



Jakarta EE 9 Web Profile

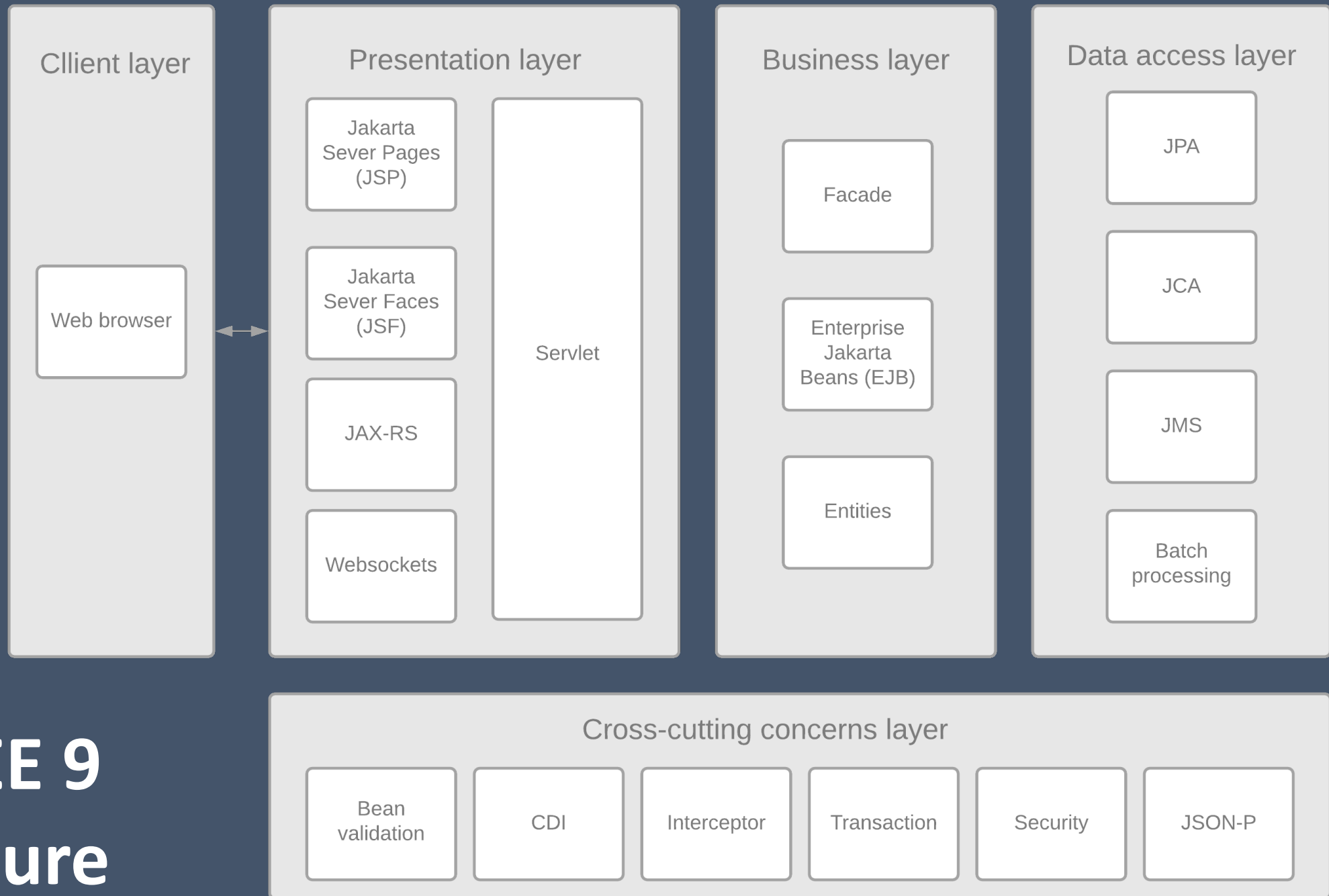
Typical Jakarta EE architecture

Lecture: Typical Jakarta EE architecture

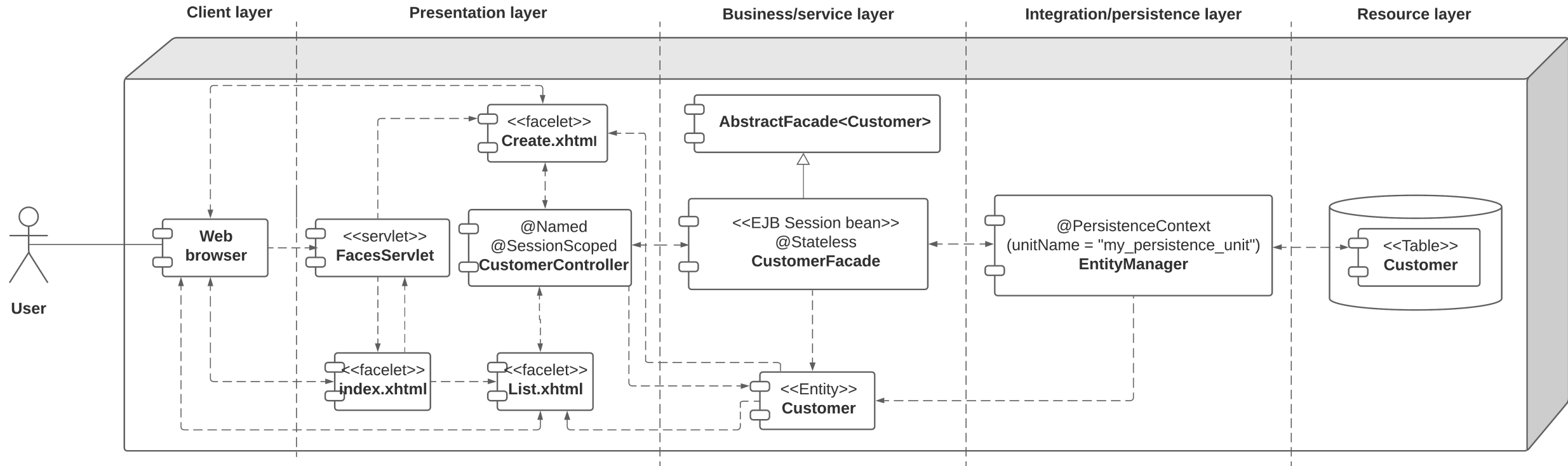
Module: Introduction to Jakarta EE

Typical Jakarta EE architecture

Typical Jakarta EE 9 architecture



Typical architecture for a simple web application with Jakarta EE

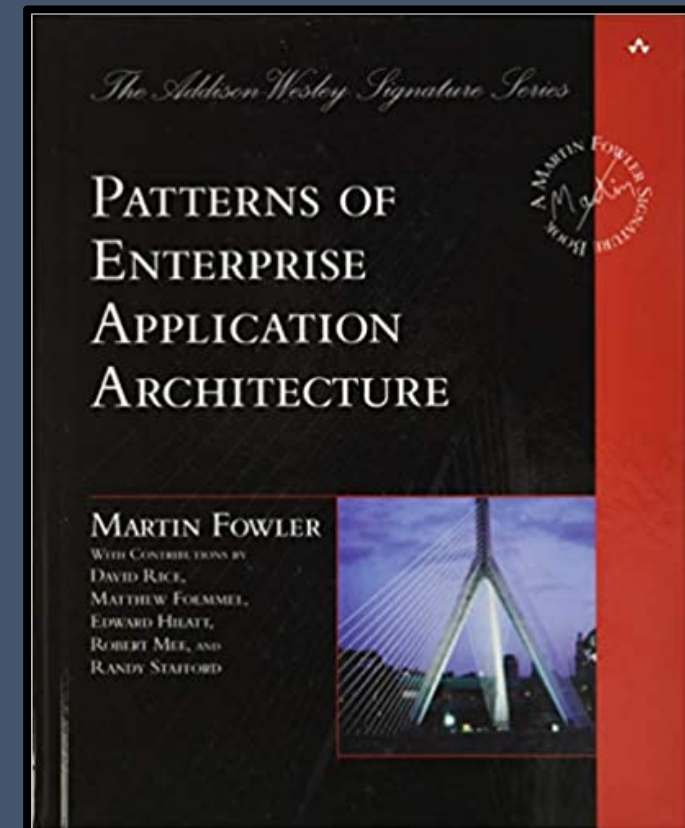
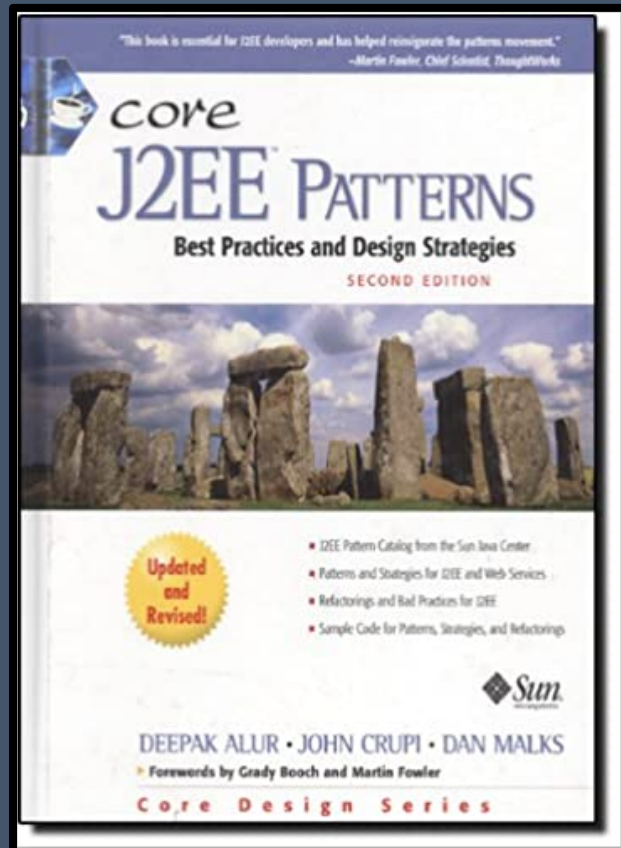


Recommended reading

Lecture: Recommended reading

Module: Jakarta EE

Recommended reading



Questions?